

[illegible]

Mol Fraction B	0.7	
Mol Fraction T	0.5	
Pressure (kpa)	101.32	
Pressure (Atm)	1	
Feed Vaporized	1/3	
L1 (mols)	100	Amount starting with
L2 (mols)	66.666	Amount ending with
L1/L2	1.50015	
ln(L1/L2)	0.4054751082	
Alpha	2.4	From PS03
X	Y	1/y-x
0.025	0.05797101449	30.32967033
0.05	0.1121495327	16.09022556
0.075	0.1628959276	11.37709138
0.1	0.2105263158	9.047619048
0.125	0.25531911489	7.673469388
0.15	0.2975206612	6.778711485
0.175	0.3373493976	6.159554731
0.2	0.375	5.714285714
0.225	0.4106463878	5.386584741
0.25	0.4444444444	5.142857143
0.275	0.476534296	4.961934617
0.3	0.5070422535	4.829931973
0.325	0.5360824742	4.737484737
0.35	0.5637583893	4.678178964
0.375	0.5901639344	4.647619048
0.4	0.6153846154	4.642857143
0.425	0.6394984326	4.662038729
0.45	0.6625766871	4.704184704
0.475	0.6846846847	4.769065521
0.5	0.7058823529	4.857142857
0.525	0.7262247839	4.96566774
0.55	0.7457627119	5.108225108
0.575	0.7645420363	5.27584947
0.6	0.7826086957	5.476190476
0.625	0.8	5.714285714
0.65	0.8167539267	5.996860283
0.675	0.8329048843	6.332926333
0.7	0.8484848485	6.734693878
0.725	0.8635235732	7.218987909
0.75	0.8780487805	7.80952381
0.775	0.8920863309	8.540706605
0.8	0.9056603774	9.464285714
0.825	0.9187935035	10.66171923
0.85	0.9315068493	12.26890756
0.875	0.9438202247	14.53061224
0.9	0.9557522124	17.93650794
0.925	0.9673202614	23.62934363
0.95	0.9785407725	35.03759398
0.975	0.9894291755	69.3040293
1	1	
Yavg	0.6458255333	

"MANUALLY" COMPUTED ANTIDERIVATIVE:
 $\int f(x) dx = F^*(x) =$

"Manual" integration with steps:
The calculator finds an antiderivative in a comprehensible way. Note that due to some simplifications, it might only be valid for parts of the function.

$$x \cdot (704100x^6 - 2354870x^5 + 3152688x^4 - 2158275x^3 + 805280x^2 - 162960x + 19026) + C_2$$

Show steps

X 2	0.6492063369
Y avg	0.4054752042
Check	0.0000
Finding Yav	0.8016083264
Checking w/ alternative equation	
X2	0.6316909685
Yavg	0.4054753056
Check	0.0000
Finding Yav	0.8366390632

$$\ln \frac{L_1}{L_2} = \frac{1}{2-1} \left(\ln \frac{X_1}{X_2} + 2 \ln \frac{1-X_2}{1-X_1} \right)$$

